Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5.	(a)			Electrical energy is transferred from the power station to transformer A, which is a step-up transformer. It increases the voltage and reduces the current before passing it along the cables of the National Grid. Low current reduces heat loss from the cables as they heat up less. This makes the transfer of electrical energy more efficient. At transformer B the voltage is decreased [and the current increased]. Transformer B is the step-down transformer. A low voltage in the homes is safer than a high voltage. **5–6 marks** Detailed description of both efficiency and safety. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3–4 marks** Detailed description of either efficiency or safety **or** limited descriptions of both. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1–2 marks** A limited description of either efficiency or safety. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*	6			6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			To get electricity / acts as a back-up source (1) When it is not sunny (or during the night) **or** when the panels don't generate enough electricity (1) **Alternative:** Electricity can be fed back [into the National Grid] (1) If it is very sunny **or** if the panels produce more electricity than is needed (1) Accept energy for electricity / sunlight is not reliable Don't accept solar panels aren't reliable		2		2		
	(c)	(i)		$20 \times 60 = 1\,200$ [A]		1		1	1	
		(ii)	I	Substitution: $230 \times 1\,200$ (**ecf**) (1) = 276 000 [W] (1) **Alternative:** Substitution: 230×60 (1) $13\,800 \times 20 = 276\,000$ [W] (1)	1	1		2	2	
			II	$\frac{276\,000\ \textbf{ecf}}{1000} = 276$ [kW]		1		1	1	
				Question 5 total	**7**	**5**	**0**	**12**	**4**	**0**